

COM668 Project

Planning for success

Planning

- Define
- Methodology
- Plan

Define

Know what you intend to do

Define

- Define the project Aim and Objectives
- Define the scope of the project

Aim

- What do you want to achieve
- The intended outcome rather than a specific action
- Not normally directly measurable
- Achieved over the long term

Achieve a First Class Honours degree

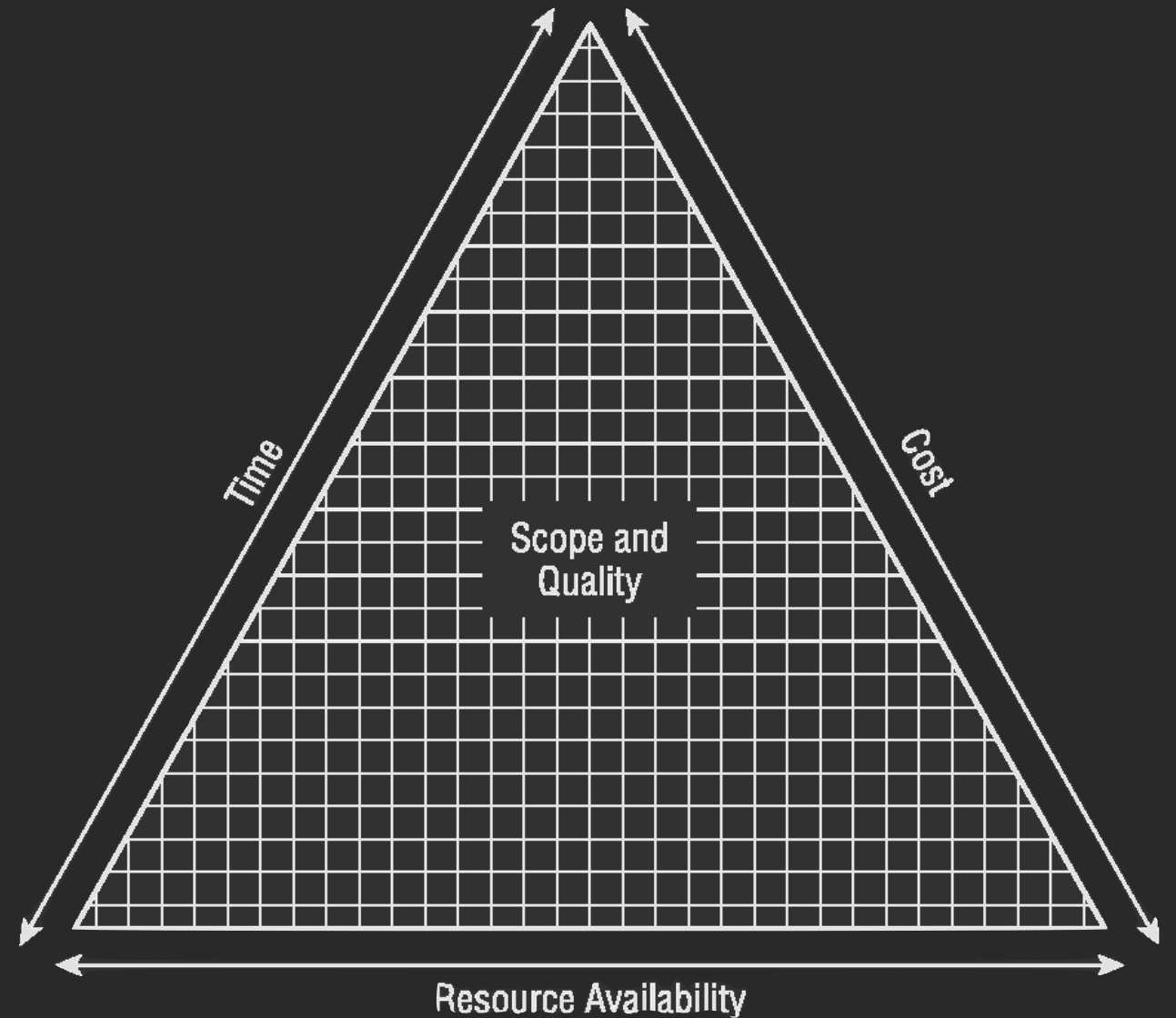
Objectives

- Attained by your actions
- Specific actions that help to attain the aim
- Measurable and tangible
- Actioned over the short to mid term

Specify the requirements for my COM668 Project on time

Scope

- Understand the interdependent nature of constraints
 - Time
 - Cost
 - Resources
 - Quality
 - Scope
- Identify what you can control
- Prioritise to meet the aim
- Plan to the constraints



Methodology

Taking a methodical approach

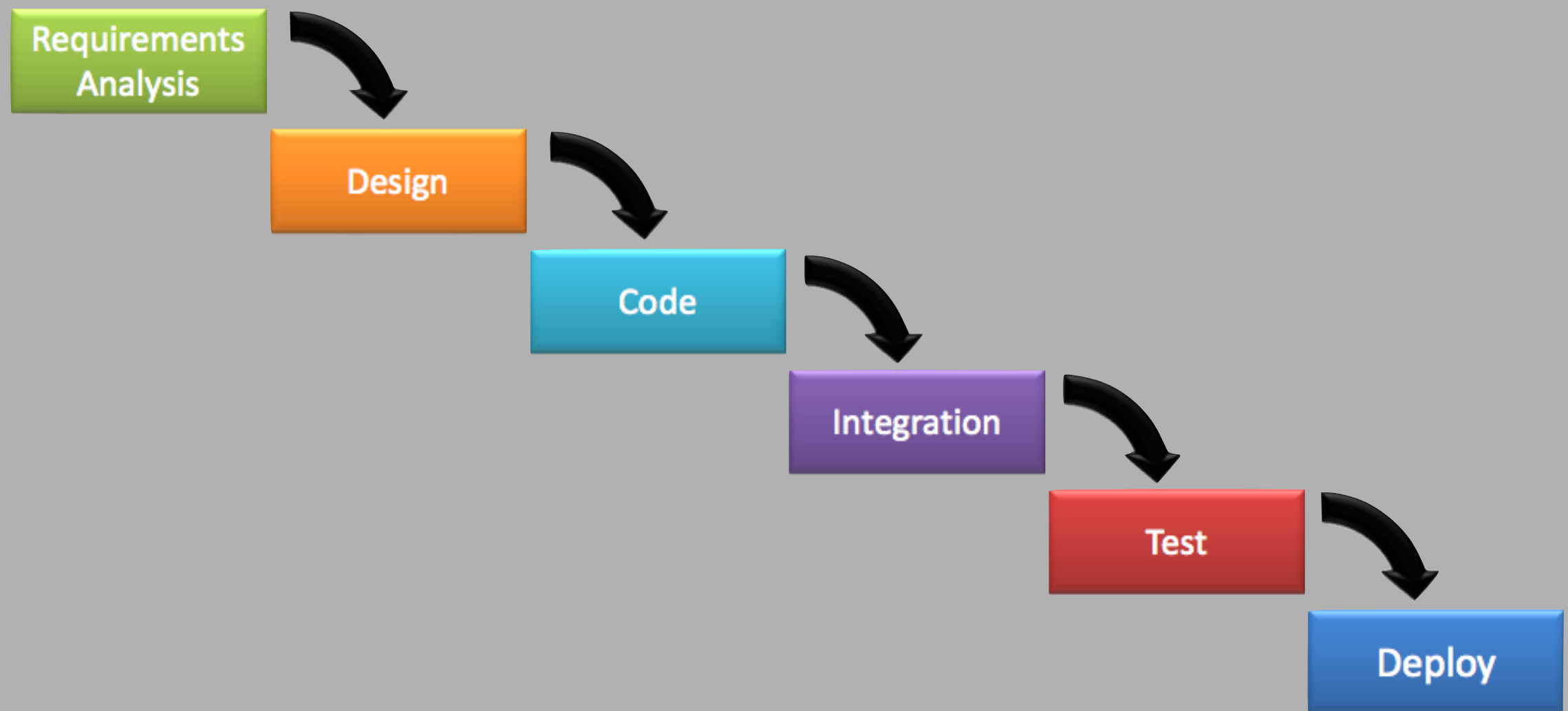
Manage

- Research available methodologies
- Select a suitable methodology
- Discuss its appropriateness

Methodologies

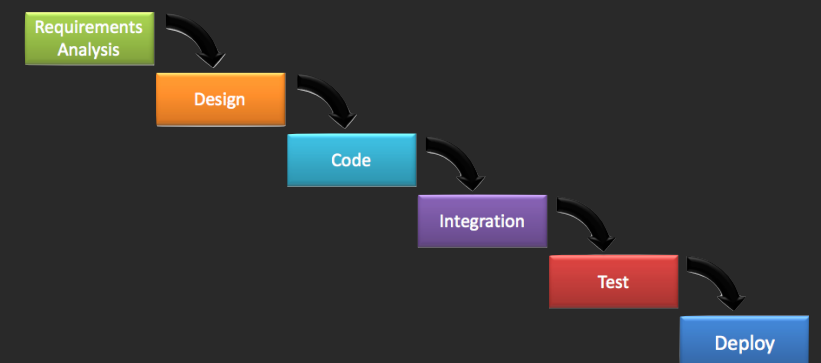
- Select a methodology to manage your activity
- A wide range exist
 - Waterfall
 - Prototyping
 - Rapid Application Development
 - Agile

Waterfall

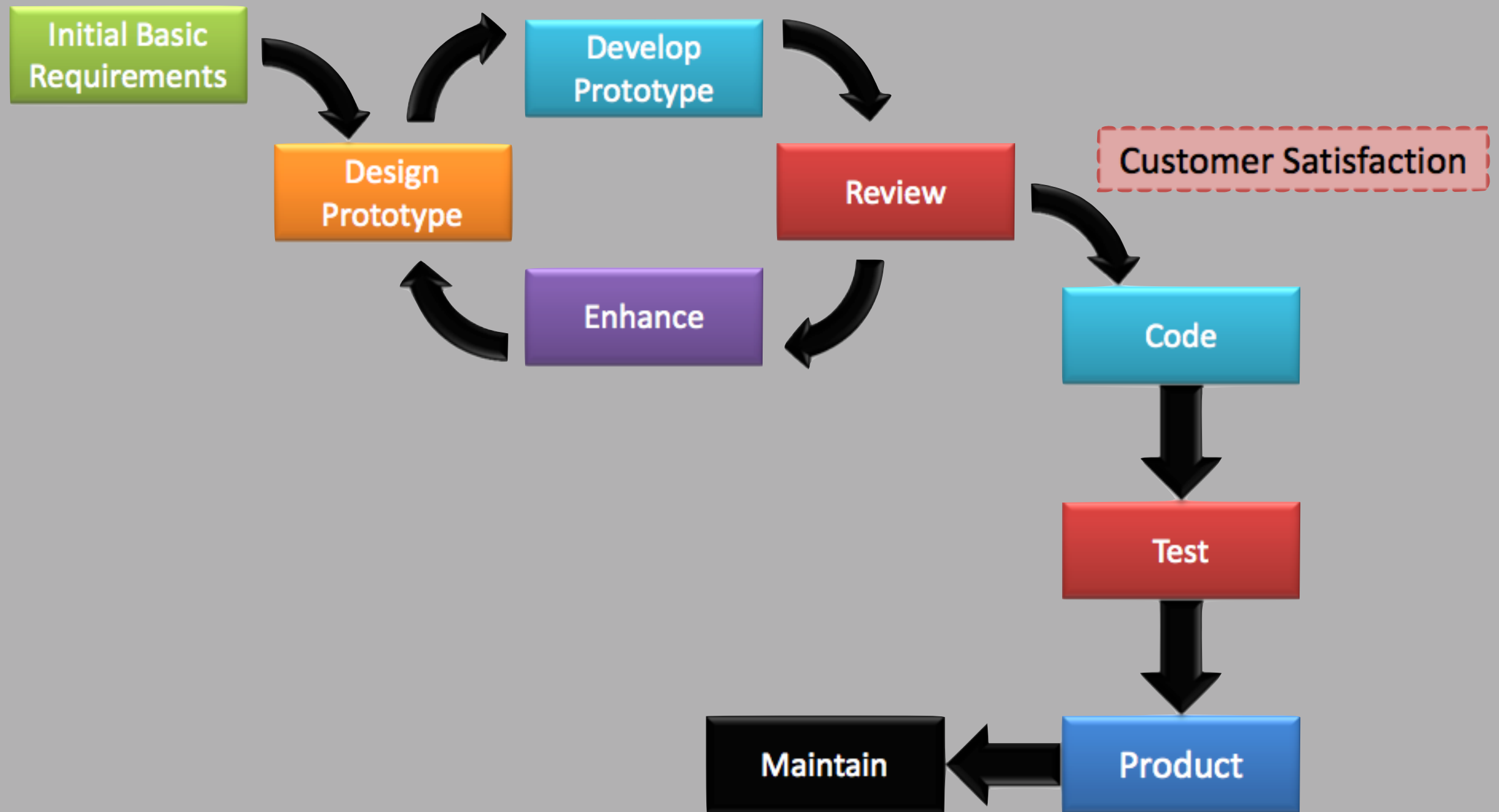


Waterfall

- A traditional methodology
- Evolved from manufacturing
- Needs well defined requirements
- Straightforward
- Easy to manage
- Rigid
- Suited to small well defined projects

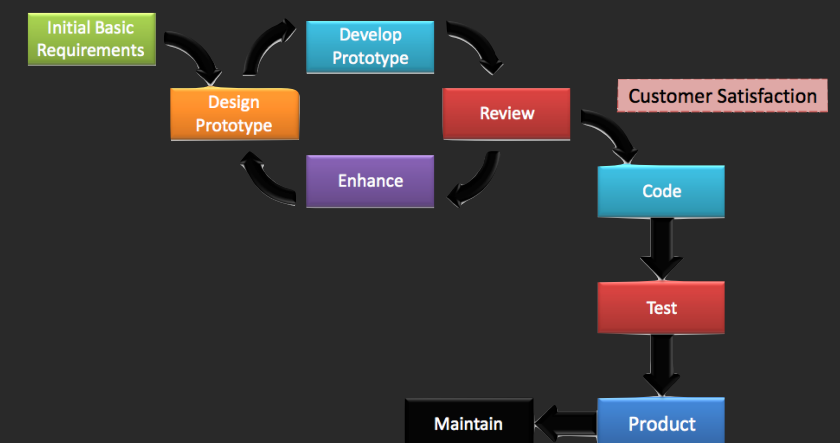


Prototyping

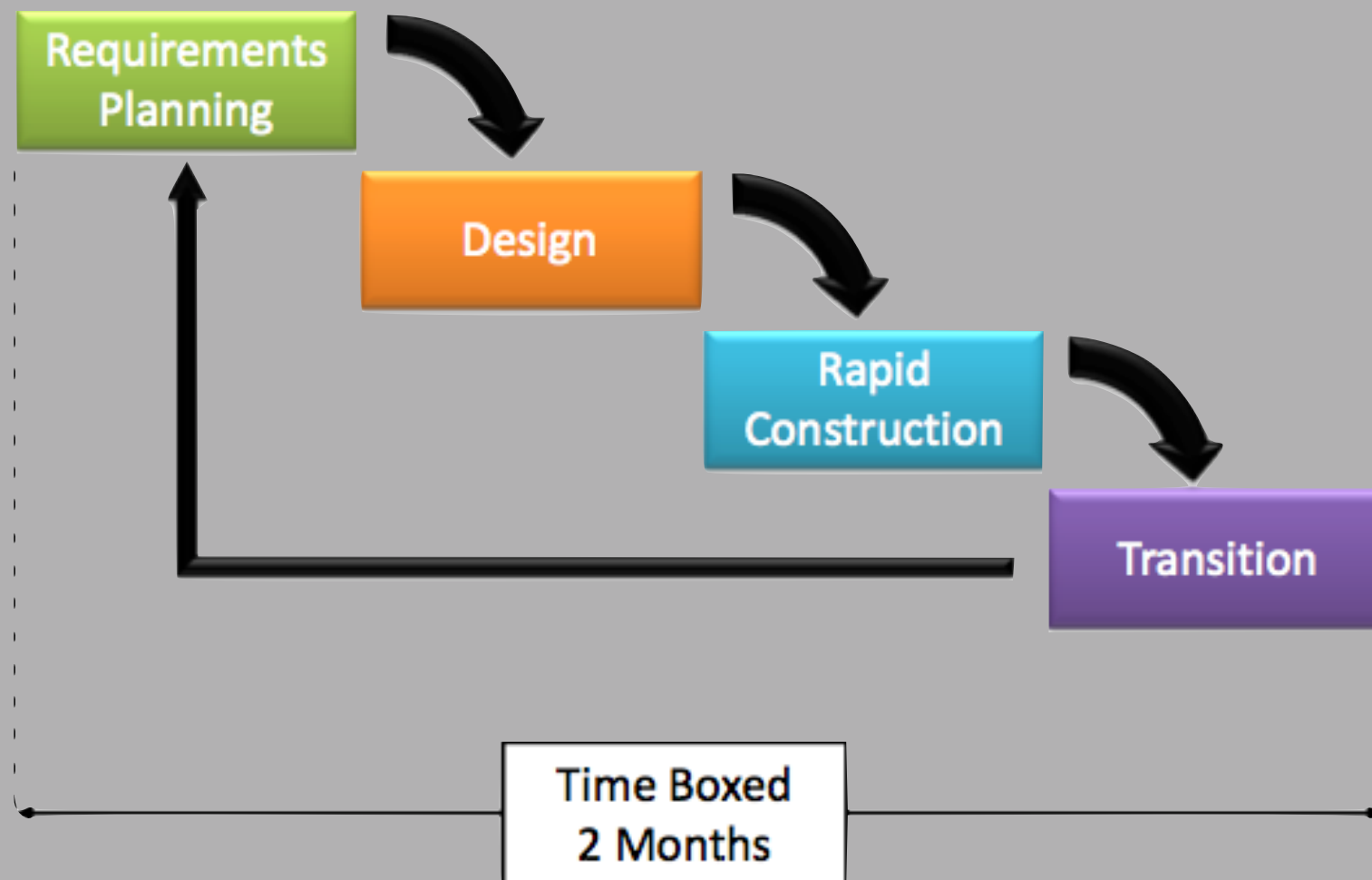


Prototyping

- Suited to projects with hard to define requirements
- Prototype helps to elicit requirements
- Quite complex
- Flexible
- Works well with user engagement
- Managing scope more challenging

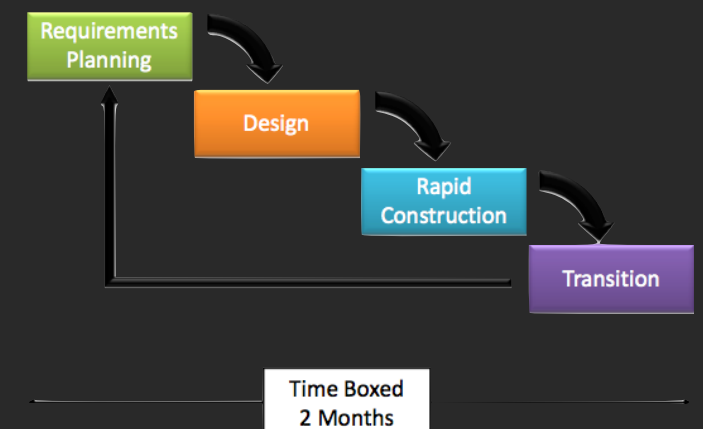


Rapid Application Development



Rapid Application Development

- Time centred approach
- Focused on delivering on time
- Other constraints subject to time



Agile

People

Results

Collaboration

Responsiveness

Agile

- Modern methodology
- Originated from the Agile manifesto
- Values
 - Individuals and interactions over processes and tools
 - Working software over comprehensive documentation
 - Customer collaboration over contract negotiation
 - Responding to change over following a plan
- Range of methods
 - Scrum
 - Story-driven
 - Extreme Programming

People

Results

Collaboration

Responsiveness

Selecting a methodology

- There are many other methodologies and variations
- Best choice depends on the project and its constraints
- Need to research and justify what is the best fit for your project

Plan

Thinking ahead and staying on track

Planning

- This can be done at a high-level for the project as a whole
- It can also be done for requirements or user stories once defined
- Provide a high-level plan
 - Tasks
 - Durations
 - Schedule

Tasks

- Identify what needs to be done so you can plan your work
- High-level approaches
 - Top-down
 - Bottom-up
 - Combination

Duration

- Define how long tasks will take to complete
- Durations can be determined by
 - Analogy
 - Expert
 - Experimentation

Schedule

- Understand how it all fits together
 - Schedule and sequence your work
 - Assign dates to the tasks
 - Identify dependencies and bottlenecks
- Visualise to help understand and manage activity
 - PERT charts can help plan your workflow
 - Gantt charts can help to manage your project
 - Burn-down charts can help track work progress vs. time
 - Which to use depends on project stage and methodology

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